OBJECT-ORIENTED ANALYSIS & DESIGN

Digital Payment Gateway Simulation

Use Case Diagram, Actors and Use Case Specifications

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202401100200100	Use Case Diagram

Task

Draw the use case diagram of the Digital Payment Gateway Simulation, specify the role of each actor, and state the precondition, post-condition and function of each use case.

1 System Overview

The gateway sits between a customer, a merchant and the banks. A customer pays by card (with an OTP) or by UPI, the gateway asks the bank to debit the customer and credit the merchant, and a receipt is issued. The merchant can raise a refund, and every step is written to a log for auditing. In the use case model the **Payment Gateway** is the system under study, so it is drawn as the **system boundary** and not as an actor. The actors are the people and external systems that interact with it from outside.

2 Actors and Their Roles

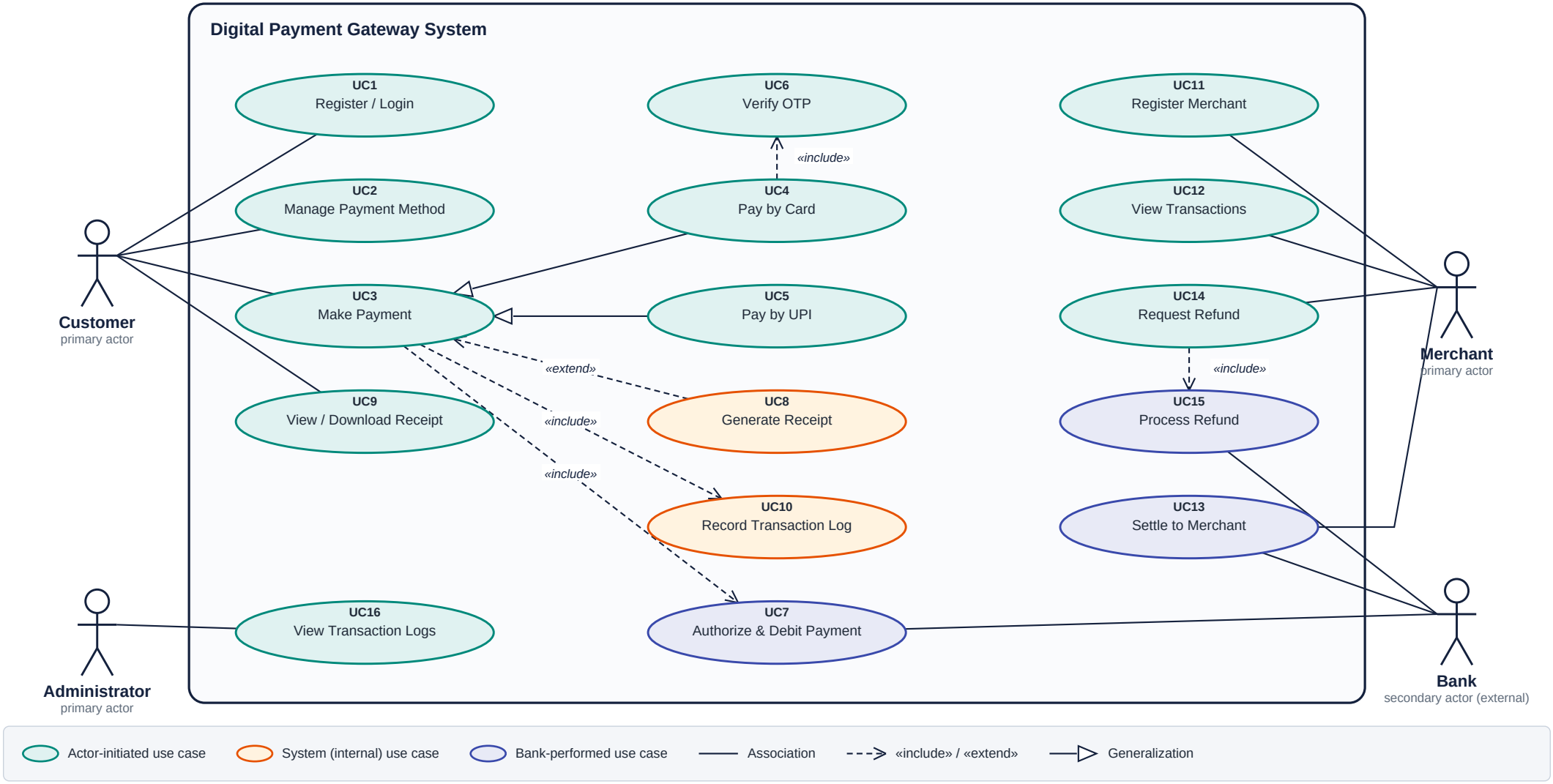
Actor	Type	Role in the system	Takes part in
Customer	Primary	Person who pays. Registers, logs in, saves a card or UPI id, starts a payment, enters the OTP or approves the UPI request, and later downloads the receipt.	UC1, UC2, UC3, UC4, UC5, UC6, UC9
Merchant	Primary	Business that receives the money. Registers with the gateway, tracks the payments it has received, receives the settlement and can raise a refund for a completed transaction.	UC11, UC12, UC13, UC14
Bank	Secondary (external)	External institution that holds the accounts. Checks available funds and debits the customer, credits the merchant at settlement, and approves refunds. It never starts a use case; the gateway calls it.	UC7, UC13, UC15
Administrator	Primary	Auditor of the gateway. Reads the transaction logs to trace a payment, investigate a failure or check a refund. Read-only access; cannot change a log entry.	UC16

The Administrator is an added actor for auditing; it uses the TransactionLog class from the class diagram. Generate Receipt and Record Transaction Log are performed by the gateway itself and have no human actor.

3 Use Cases Identified

ID	Use case	Actor(s)	Relationship
UC1	Register / Login	Customer	-
UC2	Manage Payment Method	Customer	-
UC3	Make Payment	Customer	Central use case; parent of UC4 and UC5
UC4	Pay by Card	Customer	Generalizes (is-a) UC3; includes UC6
UC5	Pay by UPI	Customer	Generalizes (is-a) UC3
UC6	Verify OTP	Customer	Included by UC4
UC7	Authorize & Debit Payment	Bank	Included by UC3
UC8	Generate Receipt	System (gateway)	Extends UC3 (only when payment succeeds)
UC9	View / Download Receipt	Customer	-
UC10	Record Transaction Log	System (gateway)	Included by UC3
UC11	Register Merchant	Merchant	-
UC12	View Transactions	Merchant	-
UC13	Settle to Merchant	Bank, Merchant	-
UC14	Request Refund	Merchant	Includes UC15
UC15	Process Refund	Bank	Included by UC14
UC16	View Transaction Logs	Administrator	-

4 Use Case Diagram - Digital Payment Gateway Simulation



5 Relationships Used in the Diagram

- **Association (solid line):** an actor takes part in a use case, e.g. Customer with Make Payment, Bank with Process Refund.
- **Include (dashed arrow to the included use case):** the base use case always runs the included one. Make Payment includes Authorize & Debit Payment and Record Transaction Log; Pay by Card includes Verify OTP; Request Refund includes Process Refund.
- **Extend (dashed arrow to the base use case):** optional behaviour added only on a condition. Generate Receipt extends Make Payment and runs only if the payment succeeds.
- **Generalization (hollow triangle):** Pay by Card and Pay by UPI are special kinds of Make Payment and inherit its actor and steps, matching the PaymentMethod inheritance in the class diagram.

6 Use Case Specifications

Each use case below lists its function, what must be true before it starts (precondition) and what is true after it finishes (post-condition).

Use case	Function	Precondition	Post-condition
UC1 **Register / Login** Actor: Customer	A new customer creates an account with name, email and phone; a registered customer logs in to the gateway.	For registration: valid email and phone. For login: the customer is already registered.	Account is created, or a login session is active. The customer can now start payments.
UC2 **Manage Payment Method** Actor: Customer	Customer adds, edits or removes a card or UPI id. The gateway validates the details before saving.	Customer is logged in.	A validated payment method is stored against the customer and can be chosen for later payments.
UC3 **Make Payment** Actor: Customer	Core use case. The customer picks the merchant, amount and a saved payment method; the gateway creates a Transaction (status Initiated), verifies it, has the bank debit it and returns the result.	Customer is logged in, has a valid payment method, and the merchant is registered.	Transaction ends as Success or Failed and the customer is shown the outcome. Every step is logged.
UC4 **Pay by Card** Actor: Customer	Validates card number, expiry date and card holder, then triggers OTP verification before the debit.	Customer selected a saved card that has not expired.	Card and OTP accepted and the request goes to the bank; otherwise the transaction is marked Failed.
UC5 **Pay by UPI** Actor: Customer	Validates the UPI id and sends a collect request to the customer's UPI app, where it is approved with the UPI PIN.	Customer selected a valid UPI id linked to a bank account.	Collect request approved and passed to the bank; if rejected or timed out the transaction is marked Failed.
UC6 **Verify OTP** Actor: Customer	Generates a one-time password for this transaction, sends it to the customer and checks the code entered before it expires.	A card payment has started and the Transaction is in Initiated state.	OTP verified and the payment continues; a wrong or expired OTP fails the transaction.
UC7 **Authorize & Debit Payment** Actor: Bank	The gateway routes the request to the bank, which checks available funds and debits the customer's account.	Card OTP verified or UPI request approved; Transaction is Initiated.	Customer account debited and status set to Success. If funds are insufficient nothing is debited and status is Failed.
UC8 **Generate Receipt** Actor: System (gateway)	Creates a Receipt with receipt number, date and amount paid for the transaction.	Transaction status is Success.	Receipt is saved with the transaction and available to the customer.
UC9 **View / Download Receipt** Actor: Customer	Customer opens or downloads the proof of payment.	Customer is logged in and a receipt exists for a successful transaction.	Receipt is displayed or downloaded. No transaction data is changed.

Use case	Function	Precondition	Post-condition
UC10 Record Transaction Log Actor: System (gateway)	Writes an audit entry (event and time) for each step: initiated, OTP sent, debited, failed, refunded.	A transaction event has occurred for an existing Transaction.	A new TransactionLog entry is saved under that transaction.
UC11 Register Merchant Actor: Merchant	A business registers with the gateway giving its business name and settlement account.	Merchant has a valid business and a bank settlement account.	A merchant id is created and the merchant can receive payments.
UC12 View Transactions Actor: Merchant	Merchant views the payments received and their status.	Merchant is registered and logged in.	Transaction list is shown. No data is changed.
UC13 Settle to Merchant Actor: Bank (beneficiary: Merchant)	After a successful payment the bank credits the merchant's settlement account.	Transaction is Success, the customer is debited and the settlement account is valid.	Merchant account is credited with the amount and the settlement is logged.
UC14 Request Refund Actor: Merchant	Merchant raises a refund for a completed transaction; the gateway creates a Refund record with amount and status.	Merchant is logged in; the transaction is Success, not already refunded, and refund amount is not more than the amount paid.	Refund is created with status Pending, passed to the bank and logged.
UC15 Process Refund Actor: Bank	Bank approves the refund, takes the amount back from the merchant and returns it to the customer.	A refund request with status Pending exists and the merchant has sufficient funds.	Money is returned to the customer and the refund is marked Approved (or Rejected). The event is logged.
UC16 View Transaction Logs Actor: Administrator	Administrator fetches the log entries of a transaction to audit a payment or investigate a failure.	Administrator is logged in with audit rights and the transaction has at least one log entry.	Logs are displayed. They are read-only and are not modified.

7 Conclusion

The Digital Payment Gateway Simulation has 4 actors (Customer, Merchant, Bank and Administrator) and 16 use cases. Make Payment is the central use case: it is specialised into Pay by Card and Pay by UPI, always includes Authorize & Debit Payment and Record Transaction Log, and is extended by Generate Receipt on success. The Bank is the only external system actor, and the Administrator reads the audit log created for every transaction.